SHAKYA BHATTACHARYYA

DATA SCIENTIST | DECISION SYSTEMS, PRODUCTION ML & EXPERIMENTATION

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PROFESSIONAL SUMMARY

Data scientist with a computer-science foundation and a continuous path from enterprise analytics and machine learning at Cognizant to graduate decision science and current production data-science ownership at Rock. Builds Python/SQL data pipelines, forecasting and anomaly analytics, model-driven decision workflows, and governed AI systems across operational data at scale. Strong in translating ambiguous business problems into measurable, production-ready decisions.

TECHNICAL SKILLS

Modeling & Decision Science: classification, regression, forecasting, anomaly detection, experimentation, A/B testing, uplift analysis, model evaluation, threshold optimization

Programming & Data: Python, SQL/PL-SQL, PostgreSQL, Oracle, pandas, NumPy, scikit-learn, XGBoost, TensorFlow, ETL, APIs, data modeling

Production & Analytics: FastAPI, MLflow, AWS, Azure, Git, Snowflake, BigQuery, Tableau, Power BI, monitoring, feature engineering

PROFESSIONAL EXPERIENCE

Rock Emergency Services | Rochester, NY — Feb 2026 - Present

Data Scientist - AI Systems & Decision Intelligence

- Built a production decision-intelligence platform spanning **1,679 projects and ~$22M in estimated revenue**, unifying operational, financial, workforce, communications, and field data into decision-ready signals.
- Engineered Python/SQL ingestion, transformation, and canonical-data pipelines across multiple systems, cutting reconciliation effort **50%+** and reducing key signal latency to **<5 minutes**.
- Developed forecasting, anomaly, and exception analytics for staffing, SLA exposure, project completeness, callbacks, and revenue signals; delivered **414 targeted alerts and 171 automated workflows**.
- Productionized governed LLM/agent workflows across **1,243 runs and 4,522 logged actions** with validation gates, human approval, audit trails, and reproducible action histories.

Cognizant Technology Solutions | Kolkata, India — Feb 2022 - Jun 2024

Junior Software Engineer | *Data Analytics & Machine Learning*

- Developed XGBoost, regression, forecasting, and classification models on **250K+ clinical-trial records and TB-scale enterprise data**, improving decision turnaround **20%** and supporting **$30M+ in cost avoidance**.
- Built production SQL ETL, feature-engineering, and batch-scoring pipelines, reducing recurring reporting cycles by **~90%** and improving model-data latency **20%**.
- Deployed and monitored TensorFlow models that improved detection precision **12%**; analyzed feature contributions, error patterns, and threshold trade-offs to improve prioritization.
- Partnered with engineering and business teams on data-quality validation, exception analysis, analytical documentation, and productionization of repeatable decision workflows.

ANALYTICS & CONSULTING EXPERIENCE

Simon Business School, University of Rochester | Rochester, NY — Jan 2025 - Dec 2025

Instructional Analytics Associate - Analytics Design & Application

- Led applied Python, SQL, predictive modeling, optimization, and BI labs for **100+ graduate students**, emphasizing reproducible analysis and business interpretation.
- Built Python-based grading and feedback automation that reduced evaluation time **70%** while improving consistency and turnaround.

Simon Vision Consulting | Rochester, NY — Sep 2024 - Dec 2024

Data Science Analyst - Client Engagement: City of Rochester

- Developed classification, A/B testing, and uplift models to identify outcome drivers, improving AUC by **0.07** and informing targeted resource-allocation strategy.
- Applied TF-IDF, topic modeling, clustering, and statistical profiling to **5,000+ survey responses**; built Power BI/SQL workflows that reduced recurring reporting effort **50%**.

PROJECT

Fraud Operations Analytics & Risk Decisioning | Python, SQL, scikit-learn — Aug 2025 - Dec 2025

- Built a cost-sensitive risk-scoring workflow using feature engineering, logistic regression, random forest, anomaly indicators, and threshold optimization under review-capacity constraints.
- Evaluated false-positive and workload trade-offs across thresholds, reducing **simulated** false positives **25-35%** and manual-review workload **~20%**; explicitly separated simulated outcomes from realized business impact.

EDUCATION

University of Rochester, Simon Business School | M.S. Business Analytics (STEM), GPA 3.9/4.0 — Jul 2024 - Dec 2025
Dean's List | Merit Scholarship | Advanced Certificate in Pricing Analytics

Maulana Abul Kalam Azad University of Technology | B.Tech. Computer Science & Engineering — Aug 2018 - Jun 2022